

```
46     dimension = input.get_shape().ndims - 2
47     base = 1.0 - tf.eye(M)
48     if dimension < 2: return tf.constant(1.0) # Identity.
49     elif dimension == 2: return base # Diagonal excluded.
50     elif dimension == 3: return tf.expand_dims(base, 2) *
        ↪ tf.expand_dims(base, 1) * tf.expand_dims(base, 0) # Mask
        ↪ out all tuples (x, y, z) that x == y or y == z or z == x.
51     else: raise NotImplementedError()
52
53 def _input_permutations(input):
54     dimension = input.get_shape().ndims - 2
55     if dimension < 2: return input
56     else: return tf.concat([
57         tf.transpose(input, [0] + list(perm) + [1 + dimension])
58         for perm in permutations(range(1, 1 + dimension))
59     ], -1)
```
